

2018 Maths First Round ESSLCE Online Exam



Question 1

Consider the data: 11, 13, 15, 17, 19, 21. Which one of the following is the mean deviation from the mean?

- a. 4
- b. 3.5
- c. 4.5
- d. 3

Question 2

Consider the quadrilateral ABCD where $A=(0,0)$, $B=(-1,1)$, $C=(0,2)$ and $D=(2,2)$. Which one of the following is true?

- a. CD is perpendicular to AB.
- b. BC is parallel to AD.
- c. CD is perpendicular to BE.
- d. AB is parallel to CD.

Question 3

Consider the greatest integer function defined by $f(x)=\lfloor x-1 \rfloor$. Which of the following sets is the domain of f ?

- a. The set of rational numbers.
- b. The set of integers.
- c. The set of real numbers.
- d. The set of natural numbers.

Question 4

Which one of the following relations is a function?

- a. $R=\{(x,y):x,y\in\mathbb{R} \text{ and } y^2=x\}$
- b. $R=\{(x,y):x,y\in\mathbb{R} \text{ and } |y|=x\}$
- c. $R=\{(x,y):x,y\in\mathbb{R} \text{ and } y=x^2\}$

d. $R=\{(x,y):x,y\in\mathbb{R} \text{ and } y^4=x\}$

Question 5

Which one of the following is a source of cash loan?

- a. Non-government organization to alleviate public problems.
- b. Investors involved in home construction for a community.
- c. Group of investors committed to drilling out underground water.
- d. A money line group investing in health improvement sectors.

Question 6

What is the average rate of change of $y=f(x)$ with respect to x on the interval $[a,b]$?

- a. $\frac{f(b)+f(a)}{2}$
- b. $\frac{b-a}{f(b)-f(a)}$
- c. $\frac{f(b)-f(a)}{2}$
- d. $\frac{f(b)-f(a)}{b-a}$

Question 7

Which one of the following sets is the solution of $\log_3 x + \log_3(x-2) = 1$?

- a. $\{-1,3\}$
- b. $\{3\}$
- c. $\{2\}$
- d. $\{0,2\}$

Question 8

Given the functions $f(x)=ax^2$ and $g(x)=ax^2+c$, for $a<0$ and $c>0$. Which of the following statements is true about the graph of f and g ?

- a. The graph of f is an upward parabola, and the graph of g is obtained from that of f by shifting it downward by cc units.
- b. The graph of f is a downward parabola, and the graph of g is obtained from that of f by shifting it to the right by cc units.
- c. The graph of f is an upward parabola, and the graph of g is obtained from that of f by shifting it upward by cc units.
- d. The graph of f is a downward parabola, and the graph of g is obtained from that of f by shifting it upward by cc units.

Question 9

Consider the test results of 20 students in a given class.

13 14 14 15 19 18 13 15 16 16

13 14 14 15 19 18 13 15 16 16

17 18 15 13 14 16 14 18 16 16

Which one of the following numbers is the mode of the above data?

- a. 16
- b. 15
- c. 14
- d. 18

Question 10

Consider the figure given below:



Which one of the following vectors is equal to $\vec{AB} + \vec{CD} + \vec{BC}$?

- a. $\vec{CD}$
- b. $\vec{AC}$
- c. $\vec{AD}$
- d. $\vec{BD}$

Question 11

Which one of the following equations represents a line that passes through the points (2, -1) and (-3, 4)?

- a. $x+y-1=0$
- b. $x+y+1=0$
- c. $x-y-1=0$
- d. $x-y+1=0$

Question 12

What is the coefficient of x^3y^4 in the expansion of $(x-2y)^7$

- a. 35
- b. 70
- c. 560
- d. -280

Question 13

Which one of the following is **NOT** a power function?

- a. $f(x)=x^3$
- b. $f(x)=3x$
- c. $f(x)=4x^{-6}$
- d. $f(x)=7x^{-1/3}$

Question 14

Given the problem: Maximize $Z=3x+y$ subject to:

$$\begin{cases} x + y \geq 2 \\ y + 2x \leq 4 \\ y \geq 1, x \geq 0 \end{cases}$$

Which one of the following is the maximum value of Z ?

- a. 2
- b. 5.5
- c. 4
- d. 5.8

Question 15

Let r_1 and r_2 be the roots of the quadratic equation $2x^2+3x-4=0$. Which one of the following is equal to

$$\frac{1}{r_1^2} + \frac{1}{r_2^2}?$$

- a. $3/4$
- b. $25/16$
- c. $3/16$
- d. $1/4$

Question 16

Consider the matrix

$$A = \begin{pmatrix} -1 & 2 & 1 \\ 3 & 4 & 2 \\ 1 & 1 & 5 \end{pmatrix}$$

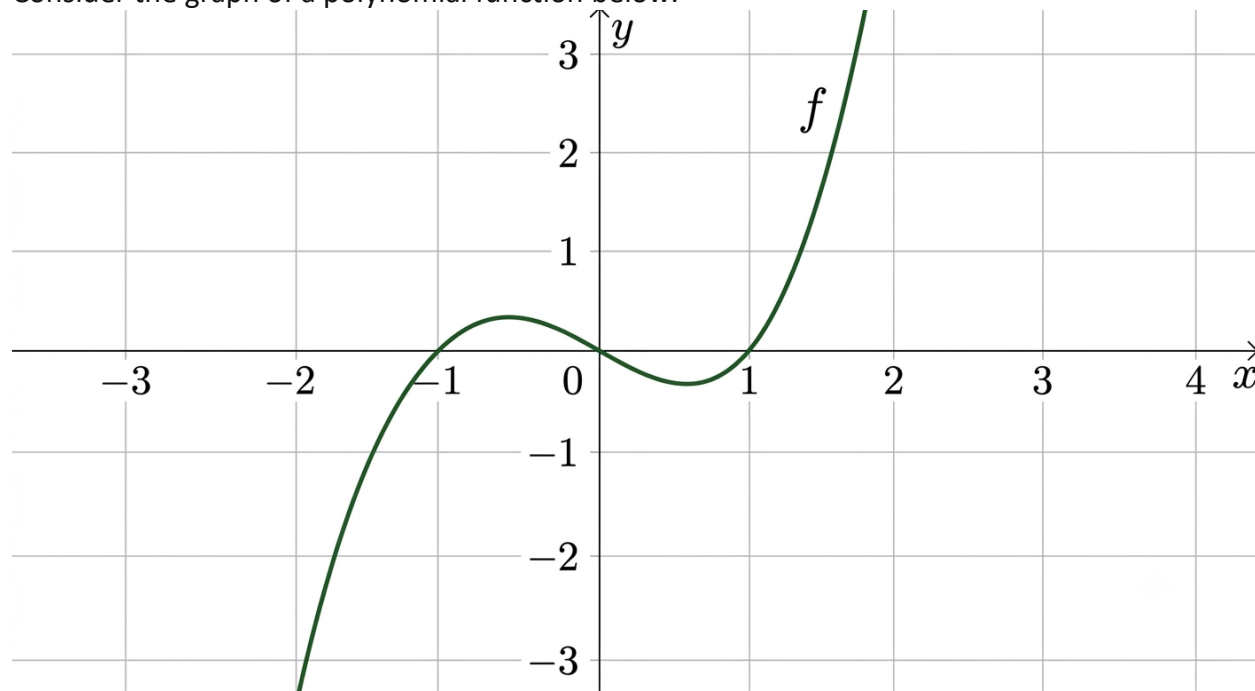
Which of the following statements is true about the minor of the entries of A ?

- a. The cofactor of 3 is 9.
- b. The cofactor of 4 is -6.

- c. The minor of 4 is 6.
- d. The minor of 3 is -9.

Question 17

Consider the graph of a polynomial function below.

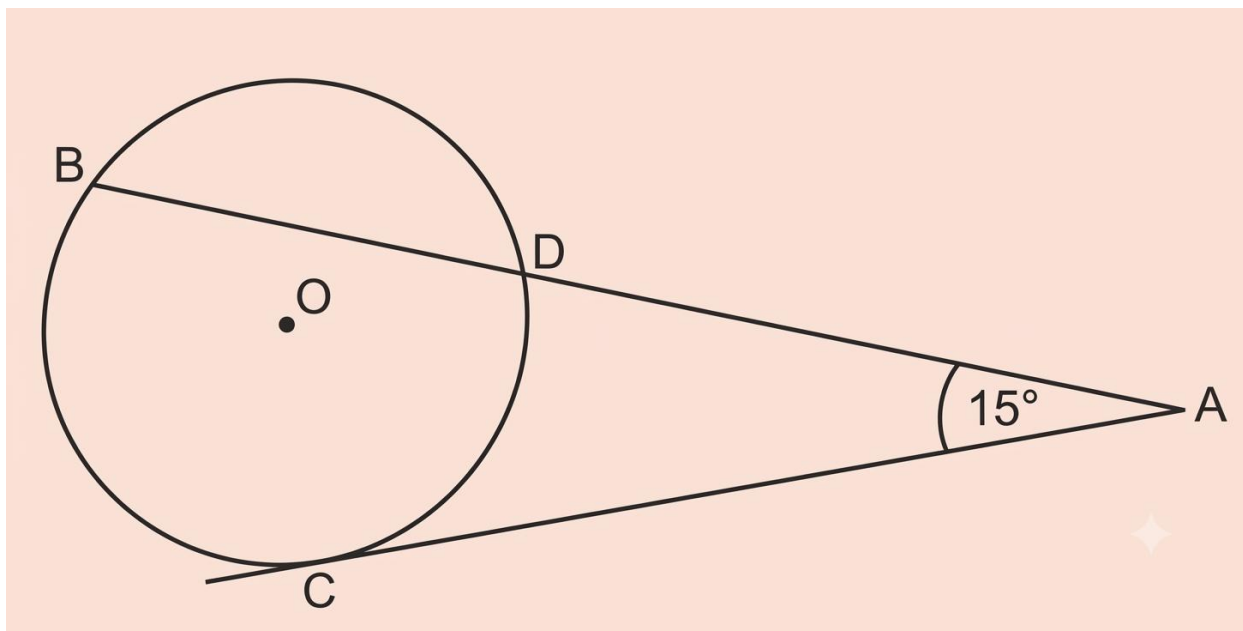


Which one of the following functions represents the above graph?

- a. $f(x)=x^3+x+1$
- b. $f(x)=x^3-x+1$
- c. $f(x)=x^3-x$

Question 18

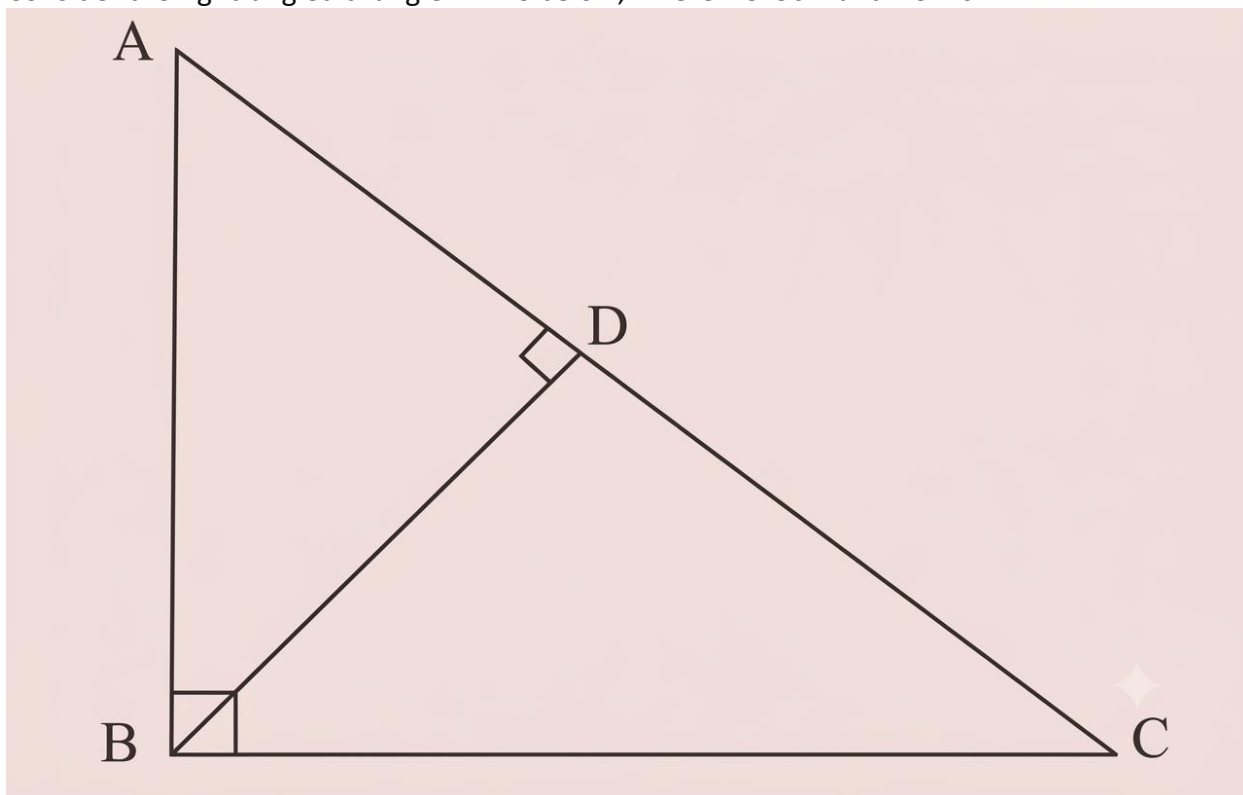
In the figure below, let $m(\angle DAC)=15^\circ$ and $m(\text{arc } BC)=70^\circ$. What is the measure of arc CD?



- a. 42.5°
- b. 55°
- c. 30°
- d. 40°

Question 19

Consider the right-angled triangle $\triangle ABC$ below, where $AC=5\text{cm}$ and $BC=4\text{cm}$.



Which one of the following is the length of BD?

- a. 5.76cm
- b. 2.4cm
- c. 5cm
- d. 7cm

Question 20

Which one of the following is true about definite integrals?

- a. $\int_a^b f(x)dx = 0, a \neq 0$
- b. $\int_a^b (f(x) + g(x))dx = \int_a^b f(x)dx + \int_a^b g(x)dx$
- c. $\int_a^b f(x)dx = \int_a^b g(x)dx$
- d. $\int_a^b (f(x) - g(x))dx = \int_a^b (g(x) - f(x))dx$

Question 21

Which of the following statements defines a circle?

- a. It is the set of points in a plane formed by the intersection of two secants in the plane.
- b. It is the set of points in a space formed by the intersection of two secants in the space.
- c. It is the set of points in a space each of which is equidistant from a fixed point in the space.
- d. It is the set of points in a plane each of which is equidistant from a fixed point in the plane.

Question 22

What is the transpose of the matrix

$$M = \begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & 3 \\ 2 & 1 & 5 \end{pmatrix}?$$

- A. $\begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 1 \\ 1 & 3 & 5 \end{pmatrix}$
- B. $\begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 3 \\ 0 & 1 & 5 \end{pmatrix}$
- C. $\begin{pmatrix} 2 & 1 & 5 \\ 0 & -1 & 3 \\ 1 & 2 & 1 \end{pmatrix}$
- D. $\begin{pmatrix} 1 & 2 & 2 \\ 0 & -1 & 1 \\ 1 & 3 & 5 \end{pmatrix}$

Question 23

Which one of the following sets is the domain of the rational function

$$f(x) = \frac{x^3 + x^2}{x^4 + 2x^2 - 3}?$$

- A. $\mathbb{R} \setminus \{1\}$
- B. $\mathbb{R} \setminus \{-1, 1\}$
- C. $\mathbb{R} \setminus \{-3, 1\}$
- D. $\mathbb{R} \setminus \{-1, 1, -3\}$

Question 24

Which of the following statements is true in borrowing money?

- A. The borrower only pays back the principal amount.
- B. The lender assumes no risk when giving out a loan.
- C. Borrowing money involves an agreement to repay the principal amount along with accrued interest.
- D. Collateral is always required for any type of financial borrowing.

Question 25

Let $f(x)=2x+1/x-1$. Which of the following equations is the equation of the tangent line to the graph of f at $(2,f(2))$?

- A. $3x+y=11$
- B. $y-3x=-1$
- C. $x+3y=17$
- D. $3x-y=13$

Question 26

What is the measure of each interior angle of a regular polygon with 15 sides?

- A. 144°
- B. 150°
- C. 156°
- D. 162°

Question 27

Let $f(x)=x^3+4x$ and $g(x)=3x^3-x^2$. What is the derivative of $(f+g)(x)$?

- A. $12x^2-2x+4$
- B. $4x^3-x^2+4x$
- C. $12x^2-x+4$
- D. $9x^2-2x+4$

Question 28

Let $A=\{x \in \mathbb{R}: x^3-x=0\}$ and $B=\{x \in \mathbb{Z}: -1 \leq x < 3\}$. What is the relative complement of B with respect to A (i.e., $A \setminus B$)?

- A. $\{2\}$
- B. $\{0,1\}$
- C. $\emptyset$
- D. $\{-1,0,1\}$

Question 29

Given the set $B=\{\emptyset, \{-2,0\}\}$, what is the power set of B ?

- A. $\{\emptyset, \{-2,0\}\}$
- B. $\{\emptyset, \{\emptyset\}, \{-2,0\}, \{\emptyset, \{-2,0\}\}\}$
- C. $\{\{\emptyset\}, \{-2,0\}\}$
- D. $\{\emptyset, \{-2\}, \{0\}, \{-2,0\}\}$

Question 30

What is the simplified form of

$$\log_{16}(4) - \log_2(\sqrt{12.5}) + \log_{0.5}(2)?$$

- A. $\log_2(5)$
- B. $1 - \log_2(5)$
- C. $-\log_2(5)$
- D. 0

Question 31

Which statement is true about the graphs of the exponential functions $f(x)=a^x$ and $g(x)=(1/a)^x$ where $a>1$?

- A. They are reflections of each other across the x-axis.
- B. They are reflections of each other across the y-axis.
- C. They never intersect each other.
- D. Both functions are monotonically increasing.

Question 32

Which of the following statements is NOT true about the mean of a given data set?

- A. It is highly affected by extreme values (outliers).
- B. The sum of the deviations of the items from the mean is always zero.
- C. It must always be one of the actual data values present in the set.
- D. It uses every value in the data set for its calculation.

Question 33

Which one of the following numbers is an irrational number?

- A. 3.14159 (terminating)
- B. $\sqrt{16}$
- C. $22/7$
- D. π

Question 34

What is the area of the parallelogram spanned by the vectors $\vec{u}=i-2j+k$ and $\vec{v}=2i+2j-k$

- A. $\sqrt{14}$
- B. $3\sqrt{5}$
- C. 9
- D. $\sqrt{35}$

Question 35

What is the fundamental period of the function $f(x)=\sin(x)$?

- A. $\pi/2$
- B. π
- C. 2π
- D. 4π

Question 36

What is the simplified form of the expression

$$(\sqrt{45} + \frac{1}{\sqrt{5}})(\sqrt{5} - \sqrt{20})?$$

- A. 16
- B. -16
- C. 8
- D. -8

Question 37

In statistical sampling, if a population is divided into homogeneous groups and a random sample is drawn from each group, this method is known as:

- A. Simple random sampling
- B. Cluster sampling
- C. Stratified sampling
- D. Systematic sampling

Question 38

A mathematics exam contains 10 multiple-choice questions and 3 short-answer questions. According to the multiplication principle, how many ways can a student select exactly one multiple-choice question and exactly one short-answer question to solve first?

- A. 13
- B. 30
- C. 10^3
- D. $\binom{13}{2}$

Question 39

A translation vector maps the origin (0,0) to the point (-2,4). Under this same translation, what is the image of the point (3,1)?

- A. (5,-3)
- B. (1,5)
- C. (-6,4)
- D. (-5,3)

Question 40

If the initial point of a vector is A(2,-1,4) and the terminal point is B(4,3,2), what is the position vector $\vec{AB}$?

- A. $\langle -2, -4, 2 \rangle$
- B. $\langle 6, 2, 6 \rangle$
- C. $\langle 2, 4, -2 \rangle$
- D. $\langle 8, -3, 8 \rangle$

Question 41

Let vector $\vec{u}$ have an initial point at (1,2) and a terminal point at (-3,6). What are the components of the vector $-\frac{1}{2}\vec{u}$?

- A. $\langle 2, -2 \rangle$
- B. $\langle -2, 2 \rangle$
- C. $\langle 4, -4 \rangle$

D. $\langle 1, -1 \rangle$

Question 42

Consider the rational function $f(x) = x^2 - 4/x - 2$. The domain of f is the set of all real numbers EXCEPT:

- A. -2
- B. 2
- C. 0
- D. The domain is all real numbers.

Question 43

Which of the following properties of matrices and scalar multiplication r is TRUE for any two matrices A and B of the same dimension?

- A. $r(AB) = (rA)(rB)$
- B. $r(A+B) = rA + rB$
- C. $(A+B)^T = A^T - B^T$
- D. $\det(A+B) = \det(A) + \det(B)$

Question 44

If AA is a 3×3 square matrix, which of the following is always true about its transpose AT ?

- A. $\det(A^T) = -\det(A)$
- B. $\det(A^T) = 1/\det(A)$
- C. $\det(A^T) = \det(A)$
- D. $A^T = A^{-1}$

Question 45

Let AA be a 3×3 invertible matrix with determinant $|A| = 4$. What is the determinant of $2(A^{-1})$?

- A. $1/2$
- B. 2
- C. 8
- D. $1/4$

Question 46

What is the solution set for the radical equation

$$\sqrt{x^2 + 3x} = 2?$$

- A. $\{1, -4\}$
- B. $\{1, 4\}$
- C. $\{-1, 4\}$
- D. $\{-1, -4\}$

Question 47

Using logarithm rules, which expression is equivalent to $\log_3(9x^2)$

- A. $2\log_3(3x)$
- B. $2 + 2\log_3(x)$
- C. $9\log_3(x^2)$
- D. $3 + 2\log_3(x)$

Question 48

Given the exponential functions $f(x)=2^x$ and $g(x)=0.5^x$, which statement describes their monotonicity?

- A. Both functions are increasing.
- B. Both functions are decreasing.
- C. $f(x)$ is decreasing and $g(x)$ is increasing.
- D. $f(x)$ is increasing and $g(x)$ is decreasing.

Question 49

In an arithmetic progression, the first term is 11 and the common difference is -13 . What is the sum of the first 7 terms?

- A. -28
- B. -196
- C. -67
- D. 196

Question 50

In a geometric sequence, the first term is $g_1=2$ and the fifth term is $g_5=8$ (assume the common ratio is positive). What is the value of the 13th term (g_{13})?

- A. 32
- B. 64
- C. 128
- D. 256

Question 51

The infinite repeating decimal $0.373737\dots$ can be expressed as which rational fraction in lowest terms?

- A. $37/100$
- B. $37/99$
- C. $37/90$
- D. $1/3$

Question 52

According to the properties of integration, which of the following is equivalent to $\int (3f(x)-g(x))dx$?

- A. $3 \int f(x)dx \cdot \int g(x)dx$
- B. $\int 3f(x)dx - \int g(x)dx$
- C. $3(\int f(x)dx - \int g(x)dx)$
- D. $\int f(x)dx - 3 \int g(x)dx$

Question 53

Three vertices of a parallelogram ABCD are $A(0,0)$, $B(5,0)$, and $C(7,4)$. If AD is parallel to BC, what are the coordinates of vertex D?

- A. (2,4)
- B. (-2,4)

- C. (12,4)
- D. (2,-4)

Question 54

If α and β are complementary angles ($\alpha + \beta = 90^\circ$), which of the following trigonometric relations is always true?

- A. $\sin(\alpha) = \sin(\beta)$
- B. $\tan(\alpha) = \tan(\beta)$
- C. $\sin(\alpha) = \cos(\beta)$
- D. $\sec(\alpha) = \cos(\beta)$

Question 55

In descriptive statistics, the third decile (D_3) corresponds to which percentile?

- A. 3rd percentile
- B. 30th percentile
- C. 33rd percentile
- D. 70th percentile

Question 56

Which one of the following statements defines a matrix?

- A. A matrix is a rectangular array of numbers.
- B. A matrix is a vertical array of letters.
- C. A matrix is a horizontal array of letters.
- D. A matrix is a circular array of numbers.

Question 57

Which one of the following is the decomposition of the rational expression $3x^2 + 5x - 1 / (x+2)(x+1)^2$ as a sum of partial fractions?

- A. $\frac{2}{x+2} + \frac{3}{x+1} + \frac{1}{(x+1)^2}$
- B. $\frac{3}{x+2} - \frac{3}{(x+1)^2}$
- C. $\frac{1}{x+2} + \frac{2}{x+1} - \frac{3}{(x+1)^2}$
- D. $\frac{5}{x+2} + \frac{4}{(x+1)^2}$

Question 58

A farmer having a total capital of Birr 400,000 bought oxen and sheep. An ox costs Birr 60,000 and a sheep Birr 5,000. He sold each ox at Birr 65,000 and each sheep at Birr 6,000. Which one of the following is a representation of the problem in order to maximize his profit, assuming uniform prices where x is the number of oxen and y is the number of sheep?

- A. $Z(x,y) = 6000x + 5000y$ subject to $x + y \leq 35, 12x + y \leq 80, x \geq 0, y \geq 0$
- B. $Z(x,y) = 60000x + 5000y$ subject to $x + y \leq 35, 12x + y \leq 80, x \geq 0, y \geq 0$
- C. $Z(x,y) = 5000x + 1000y$ subject to $x + y \leq 35, 12x + y \leq 800, x \geq 0, y \geq 0$
- D. $Z(x,y) = 5000x + 1000y$ subject to $x + y \leq 35, 12x + y \leq 80, x \geq 0, y \geq 0$

Question 59

Which one of the following statements is true about the given numbers and points on the number line?

- A. There is a one-to-one correspondence between the set of natural numbers and points on the number line.
- B. There is a one-to-one correspondence between the set of rational numbers and points on the number line.
- C. There is a one-to-one correspondence between the set of irrational numbers and points on the number line.
- D. There is a one-to-one correspondence between the set of real numbers and points on the number line.

Question 60

A spherical ball has diameter 30cm. What is its surface area (in cm^2)?

- A. 3600π
- B. 1000π
- C. 900π
- D. 600π

ANSWER KEY

- 1. d
- 2. b
- 3. c
- 4. c
- 5. d
- 6. d
- 7. b
- 8. d
- 9. a
- 10. c
- 11. a
- 12. c
- 13. c
- 14. c
- 15. b
- 16. d
- 17. c
- 18. d
- 19. b
- 20. b
- 21. d
- 22. B
- 23. B

- 24. C
- 25. A
- 26. C
- 27. A
- 28. C
- 29. D
- 30. C
- 31. B
- 32. C
- 33. D
- 34. B
- 35. C
- 36. B
- 37. C
- 38. B
- 39. B
- 40. C
- 41. A
- 42. B
- 43. B
- 44. C
- 45. B
- 46. A
- 47. B
- 48. D
- 49. B
- 50. C
- 51. B
- 52. A
- 53. A
- 54. C
- 55. B
- 56. A
- 57. A
- 58. D
- 59. D
- 60. C



THE END
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